

# Decoupled PRIMITIVE Dense Reward GUIDANCE (DPDRG) sounds cool enough

## THINGS TO RE-READ:

- **Feynman-Kac steering**
- **GLASS flows** - an ODE based way to sample transitions by simulating “inner” flow matching model derived from a pretrained model WITHOUT TRAINING, aiming to remove the common stochasticity–efficiency tradeoff and serving as a drop-in improvement for reward alignment procedures
- SDEs and why they are markedly less efficient than ODEs
- **Dense GRPO** and the problem with sparse rewards
- Semantic Credit assignment and **GenEval2** Findings
- THE BIGGEST PROBLEMS THAT INTERACT WITH REWARD HACKING
  - OPTIMIZING A PROXY SCORE DEGRADES TRUE ALIGNMENT. (**GARDO PAPER**)
  - TRAINING FREE INFERENCE TIME ALIGNMENT - HACKING SHOWS UP WHEN NOISE OPTIMIZATION DONE - NOISE SPACE REGULARIZATION IS INEFFICIENT AND AN EXPLICIT IMAGE SPACE CONSTRAINT HELPS (**MIRA Paper**)

## Area we propose targeting:

- Simultaneously solving this problem via:
  - Dense reward over timesteps
  - Decomposed prompt primitives
  - Inference time steering or lightweight post training

## HOW DO THIS? I GOT NO FUCKING CLUE WOOOO YEAH BABY

## WHAT REMAINS UNSOLVED?

- **(DENSE GRPO)** Dense reward be good - no work done on not having the reward being scalar anymore.
- **(BRANCH GRPO)** Restructuring diffusion roll outs into a tree - cheapens stuff will improving performance because the sharing prefixes but densifying the rewards - semantic stuff still not targetted
- **(Glass flows)** GLASS constructs ODE-simulated transitions from pretrained models without retraining and reports improved performance when combined with FK steering
- **(TIFA + GENEVAL2)** TIFA BUSTS OUT INTERPRETABLE VQA QA PAIRS TO SEE IF MODELS ACTUALLY SEE WHAT THEY SAY; SHOWS MODELS ARE OK AT ATTRIBUTES BUT CRAP AT SPATIAL/COUNTING. GENEVAL 2 SAYS STATIC JUDGES DRIFT AND USES PRIMITIVE TEMPLATES + SOFT-TIFA TO PUNISH MISSING ATOMS VIA GEOMETRIC MEAN.

- Meanwhile, prompt/time decomposition and stage-scheduling ideas show that **different semantics “want” different denoising stages**:
  - **STAGE-AWARE PROMPTING (SAP)** USES AN LLM TO CHOP PROMPTS INTO PROXIES FOR SPECIFIC TIMESTEP INTERVALS BECAUSE DENOISING IS COARSE-TO-FINE.
  - **DIT-ST** SMASHES CAPTIONS INTO PRIMITIVE SENTENCES AND INJECTS THEM HIERARCHICALLY ACROSS STAGES BASED ON TYPE SENSITIVITY.
  - **SPLITFLOW** DECOMPOSES PROMPTS INTO SUB-PROMPTS, RUNS INDEPENDENT FLOWS, AND AGGREGATES THEM TO STOP SEMANTIC CONFLICTS DURING EDITING.

## PROJECT THEN?

### Curr idea as i type this bullshit

- Which prompt primitives are being satisfied at which denoising steps, and can we use that signal to steer generation better
- In simpler terms:
  - break the prompt into parts
  - track those parts during generation
  - reward the model for making progress on each part
  - guide sampling using that finer signal

So instead of:

- “final image got 0.76 reward”

you get something closer to:

- object A became correct at mid steps
- color binding improved late
- count stayed wrong throughout
- spatial relation regressed after step 18
- **claim**: Primitive-dense, stepwise rewards improve compositional faithfulness better than terminal scalar rewards or dense scalar rewards.
  - PREMISES THEN:
    - Dense reward over timesteps better than terminal reward
    - Primitive level reward better than one holistic reward
    - Different primitives matter at different moments of image
    - Primitive awareness be good to stop reward hacking

## PIPELINE:

- PROMPT DECOMPOSITION
- PRIMITIVE JUDGE -TIFA STYLE OR DIRECT VLM VERIFY ER
- INTERMEDIATE IMAGE RECOVERY - AT SELECTED TIME STEPS DECODE THE CURR LATENT AND APPROX THE X0 IMAGE
- PER PRIMITIVE STEPWISE SCORING
- DO REWARD GAIN TO MEASURE PRIMITIVE PROGRESS - REWARD REFLECTS WHAT STEP THEN IMPROVED HOW AND HOW MUCH
- COMBINE THE PRIMITIVE REWARDS - FIGURE OUT HOW TO AGGREGATE (WEIGHTED OR SOFT MIN OR GEOMETRIC OR GROUP DRO STYLE)
- DO ALIGNMENT VIA REWARD - INFERENCE TIME GUIDANCE FOR NOW - MOVE TO TRAINING IF WE GO BIG ON THE PROJECT

## PLAN THEN?

- FROZEN GENERATOR + INFERENCE TIME STEERING
  - COMPACT OPEN TEXT TO IMAGE MODEL
  - FROZEN JUDGE
  - PRIMITIVE DECOMPOSITION
  - STEPWISE SCORE OVER SUBSET TIME STEPS
  - REWARD GUIDED INFERENCE TIME SAMPLING

## THINGS TO ANSWER

- Does primitive-dense reward outperform terminal scalar reward on compositional prompts?
- Does primitive-dense reward outperform dense scalar reward?
  - need to show: terminal scalar < dense scalar < dense primitive
- Which primitive categories benefit most?
- At which timesteps are different primitives most steerable?
- Does primitive-dense reward reduce reward hacking or just move it around?

## Comparisons:

- No alignment - base model
- Terminal scalar reward
- Dense scalar reward
- Primitive - dense
- Best of N to show we beat all at all times

Use compositional prompts. Potential metrics

- Primitive level success
- Full prompt success
- TIFA style faithfulness
- Human pairwise evals (ask ur friends which image better)
- Mode collapse
- Failure analysis like if holistic reward increased but compositional correctness did not or if one prim improved but some other got worse and if early steering helps or hurts.

Some extra ablations we could do if we have time

- early / late/ all - what helps most
- Primitive types
- Aggregation methods
- Judge types

Compute story then?

- For compute light setup:
  - Stable Diffusion 1.5 class
  - 512 or 384 px
  - 20-30 denoising
  - Batch size 1
  - Primitive scoring only at selected time steps like 4-8 checkpoints per sample
  - Prompt has 3- 8 primitives
  - Judge is lightweight like clip or some small vqa

Then it will take 3 to 4 days potentially maybe to get it all done (on the labs trusty rtx 2080 ti that we will graciously steal from dr taj)

To Do:

- Baseline reproduction
- Prompt decomposition
- Primitive scoring at selected steps
- Small benchmark run
- A few ablations

Keep it to:

- 200-500 prompts

- 1 seed
- 3-4 methods
- 4-6 checked time steps